

Домашнее задание №3

Анализ трафика в Wireshark

Агаева Зейнаб Фуад кызы

2026-09-26

1. Цели и задачи работы
2. Процесс выполнения лабораторной работы
3. Анализ протоколов транспортного уровня
4. Анализ TCP Handshake
5. Выводы по проделанной работе

1. 1. Цели и задачи работы

1.1 Цель лабораторной работы

Изучить MAC-адресацию, получить практические навыки захвата и анализа сетевого трафика в Wireshark, исследовать работу протоколов **ARP**, **ICMP**, **HTTP**, **DNS**, **TCP**, **UDP** и **QUIC**, а также проанализировать процесс установки TCP-соединения.

2. 2. Процесс выполнения лабораторной работы

2.1 Определяю параметры сетевого интерфейса

```
Адаптер беспроводной локальной сети Беспроводная сеть:

DNS-суффикс подключения . . . . . :
Описание. . . . . : Intel(R) Wi-Fi 6E AX211 160MHz
Физический адрес. . . . . : F8-FE-5E-6F-7B-EC
DHCP включен. . . . . : Да
Автонастройка включена. . . . . : Да
Локальный IPv6-адрес канала . . . : fe80::da3b:4057:9ef4:1e28%7(Основной)
IPv4-адрес. . . . . : 10.90.5.42(Основной)
Маска подсети . . . . . : 255.255.255.0
Аренда получена. . . . . : 26 сентября 2026 г. 10:26:01
Срок аренды истекает. . . . . : 26 сентября 2026 г. 11:56:00
Основной шлюз. . . . . : 10.90.5.59
DHCP-сервер. . . . . : 10.90.5.59
IAID DHCPv6 . . . . . : 133758558
DUID клиента DHCPv6 . . . . . : 00-01-00-01-30-47-94-1A-10-FF-E0-21-D6-B4
DNS-серверы. . . . . : 10.90.5.59
NetBios через TCP/IP. . . . . : Включен

PS C:\work\zfagaeva>
PS C:\work\zfagaeva> ping 10.90.5.59

Обмен пакетами с 10.90.5.59 по с 32 байтами данных:
Ответ от 10.90.5.59: число байт=32 время=5мс TTL=64
Ответ от 10.90.5.59: число байт=32 время=4мс TTL=64
Ответ от 10.90.5.59: число байт=32 время=7мс TTL=64
Ответ от 10.90.5.59: число байт=32 время=6мс TTL=64

Статистика Ping для 10.90.5.59:
  Пакетов: отправлено = 4, получено = 4, потеряно = 0
    (0% потеря)
Приблизительное время приема-передачи в мс:
  Минимальное = 4мсек, Максимальное = 7 мсек, Среднее = 5 мсек
PS C:\work\zfagaeva> |
```

Рисунок 1: Определение параметров беспроводного сетевого интерфейса и проверка доступности шлюза

2.2 Анализирую ICMP Echo Request

arp or icmp						
No.	Time	Source	Destination	Protocol	Length	Info
42	16.381259	10.90.5.42	10.90.5.59	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 43)
43	16.386716	10.90.5.59	10.90.5.42	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=64 (request in 42)
44	17.383299	10.90.5.42	10.90.5.59	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 45)
45	17.388187	10.90.5.59	10.90.5.42	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=64 (request in 44)
48	18.395979	10.90.5.42	10.90.5.59	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 49)
49	18.403044	10.90.5.59	10.90.5.42	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=64 (request in 48)

> Frame 42: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{3ABB4A4F-DF76-4939-80B7-FC3752}

▼ Ethernet II, Src: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec), Dst: 72:18:c7:62:fa:43 (72:18:c7:62:fa:43)

- Destination: 72:18:c7:62:fa:43 (72:18:c7:62:fa:43)
 - 1. = LG bit: Locally administered address (this is NOT the factory default)
 - 0 = IG bit: Individual address (unicast)
- Source: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)
 - 0 = LG bit: Globally unique address (factory default)
 - 0 = IG bit: Individual address (unicast)

Type: IPv4 (0x0008)
[Stream index: 0]

▼ Internet Protocol Version 4, Src: 10.90.5.42, Dst: 10.90.5.59

- 0100 = Version: 4
- 0101 = Header Length: 20 bytes (5)
- > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 60
- Identification: 0x8a7e (35454)
- > 000. = Flags: 0x0
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 128
- Protocol: ICMP (1)
- Header Checksum: 0x0000 [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 10.90.5.42
- Destination Address: 10.90.5.59
- [Stream index: 7]

▼ Internet Control Message Protocol

- Type: Echo (ping) request (8)
- Code: 0
- Checksum: 0x4d5a [correct]
- [Checksum Status: Good]
- Identifier (BE): 1 (0x0001)
- Identifier (LE): 256 (0x0100)
- Sequence Number (BE): 1 (0x0001)
- Sequence Number (LE): 256 (0x0100)
- [Response frame: 43]
- > Data (32 bytes)

Рисунок 2: Анализ ICMP Echo Request от компьютера к шлюзу

2.3 Анализирую ICMP Echo Reply

arp or icmp						
No.	Time	Source	Destination	Protocol	Length	Info
42	16.381259	10.90.5.42	10.90.5.59	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 43)
43	16.386716	10.90.5.59	10.90.5.42	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=64 (request in 42)
44	17.383299	10.90.5.42	10.90.5.59	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 45)
45	17.388107	10.90.5.59	10.90.5.42	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=64 (request in 44)
48	18.395979	10.90.5.42	10.90.5.59	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 49)
49	18.403044	10.90.5.59	10.90.5.42	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=64 (request in 48)

> Frame 43: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{3ADB4A4F-DF76-4939-B007-FC3752}	0000	f8 fe 5e 6f 7b ec
Ethernet II, Src: 72:18:c7:62:fa:43 (72:18:c7:62:fa:43), Dst: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)	0010	00 3c 9b 4f 00 00
Destination: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)	0020	05 2a 00 00 55 5a
.....0..... = LG bit: Globally unique address (factory default)	0030	67 68 69 6a 6b 6c
.....0..... = IG bit: Individual address (unicast)	0040	77 61 62 63 64 65
Source: 72:18:c7:62:fa:43 (72:18:c7:62:fa:43)		
.....1..... = LG bit: Locally administered address (this is NOT the factory default)		
.....0..... = IG bit: Individual address (unicast)		
Type: IPv4 (0x0800)		
[Stream index: 0]		
> Internet Protocol Version 4, Src: 10.90.5.59, Dst: 10.90.5.42		
0100 = Version: 4		
....0101 = Header Length: 20 bytes (5)		
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)		
Total Length: 68		
Identification: 0x9b4f (39759)		
> 000. = Flags: 0x0		
...0 0000 0000 0000 = Fragment Offset: 0		
Time to Live: 64		
Protocol: ICMP (1)		
Header Checksum: 0xc059 [validation disabled]		
[Header checksum status: Unverified]		
Source Address: 10.90.5.59		
Destination Address: 10.90.5.42		
[Stream index: 7]		
> Internet Control Message Protocol		
Type: Echo (ping) reply (0)		
Code: 0		
Checksum: 0x555a [correct]		
[Checksum Status: Good]		
Identifier (BE): 1 (0x0001)		
Identifier (LE): 256 (0x0100)		
Sequence Number (BE): 1 (0x0001)		
Sequence Number (LE): 256 (0x0100)		
[Request frame: 42]		
[Response time: 5,457 ms]		
> Data (32 bytes)		

Рисунок 3: Анализ ICMP Echo Reply от шлюза

2.4 Исследуя ARP-запрос

arp or icmp						
No.	Time	Source	Destination	Protocol	Length	Info
42	16.381259	10.90.5.42	10.90.5.59	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 43)
43	16.386716	10.90.5.59	10.90.5.42	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=64 (request in 42)
44	17.383299	10.90.5.42	10.90.5.59	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 45)
45	17.388187	10.90.5.59	10.90.5.42	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=64 (request in 44)
48	18.395979	10.90.5.42	10.90.5.59	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 49)
49	18.403844	10.90.5.59	10.90.5.42	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=64 (request in 48)
50	19.409800	10.90.5.42	10.90.5.59	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 51)
51	19.416069	10.90.5.59	10.90.5.42	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1024, ttl=64 (request in 50)
52	21.489375	72:18:c7:62:fa:43	Intel_6f:7b:ec	ARP	42	who has 10.90.5.42? Tell 10.90.5.59
53	21.489484	Intel_6f:7b:ec	72:18:c7:62:fa:43	ARP	42	10.90.5.42 is at f8:fe:5e:6f:7b:ec

> Frame 52: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{3ABB4A4F-D7F6-4939-80B7-FC3752}	0000	f8 fe 5e 6f 7b ec
▼ Ethernet II, Src: 72:18:c7:62:fa:43 (72:18:c7:62:fa:43), Dst: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)	0010	00 00 06 04 00 01
Destination: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)	0020	00 00 00 00 00 00
... ..0. = LG bit: Globally unique address (factory default)		
... ..0. = IG bit: Individual address (unicast)		
Source: 72:18:c7:62:fa:43 (72:18:c7:62:fa:43)		
... ..1. = LG bit: Locally administered address (this is NOT the factory default)		
... ..0. = IG bit: Individual address (unicast)		
Type: ARP (0x0806)		
[Stream index: 0]		
▼ Address Resolution Protocol (request)		
Hardware type: Ethernet (1)		
Protocol type: IPv4 (0x0800)		
Hardware size: 6		
Protocol size: 4		
Opcode: request (1)		
Sender MAC address: 72:18:c7:62:fa:43 (72:18:c7:62:fa:43)		
Sender IP address: 10.90.5.59		
Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)		
Target IP address: 10.90.5.42		

Рисунок 4: Анализ ARP-запроса для адреса 10.90.5.42

2.5 Исследуя ARP-ответ

arp or icmp						
No.	Time	Source	Destination	Protocol	Length	Info
42	16.381259	10.90.5.42	10.90.5.59	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 43)
43	16.386716	10.90.5.59	10.90.5.42	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=64 (request in 42)
44	17.383299	10.90.5.42	10.90.5.59	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 45)
45	17.388107	10.90.5.59	10.90.5.42	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=64 (request in 44)
48	18.395979	10.90.5.42	10.90.5.59	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 49)
49	18.403044	10.90.5.59	10.90.5.42	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=64 (request in 48)
50	19.409000	10.90.5.42	10.90.5.59	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 51)
51	19.416069	10.90.5.59	10.90.5.42	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1024, ttl=64 (request in 50)
52	21.489375	72:18:c7:62:fa:43	Intel_6f:7b:ec	ARP	42	Who has 10.90.5.42? Tell 10.90.5.59
53	21.489404	Intel_6f:7b:ec	72:18:c7:62:fa:43	ARP	42	10.90.5.42 is at f8:fe:5e:6f:7b:ec

> Frame 53: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{3AB84AAF-DF76-4939-80B7-FC3752}	0000	72 18 c7 62 fa 43 f8 fe
▼ Ethernet II, Src: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec), Dst: 72:18:c7:62:fa:43 (72:18:c7:62:fa:43)	0010	08 00 06 04 00 02 f8 fe
▼ Destination: 72:18:c7:62:fa:43 (72:18:c7:62:fa:43)	0020	72 18 c7 62 fa 43 0a 5a
....1. = LG bit: Locally administered address (this is NOT the factory default)		
....0 = IG bit: Individual address (unicast)		
▼ Source: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)		
....0. = LG bit: Globally unique address (factory default)		
....0 = IG bit: Individual address (unicast)		
Type: ARP (0x0006)		
[Stream index: 0]		
▼ Address Resolution Protocol (reply)		
Hardware type: Ethernet (1)		
Protocol type: IPv4 (0x0008)		
Hardware size: 6		
Protocol size: 4		
Opcode: reply (2)		
Sender MAC address: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)		
Sender IP address: 10.90.5.42		
Target MAC address: 72:18:c7:62:fa:43 (72:18:c7:62:fa:43)		
Target IP address: 10.90.5.59		

Рисунок 5: Анализ ARP-ответа с MAC-адресом компьютера

2.6 Проверяю доступность удалённого узла

```
PS C:\work\zfagaeva>
PS C:\work\zfagaeva> ping yandex.ru

Обмен пакетами с yandex.ru [5.255.255.77] с 32 байтами данных:
Ответ от 5.255.255.77: число байт=32 время=47мс TTL=50
Ответ от 5.255.255.77: число байт=32 время=56мс TTL=50
Ответ от 5.255.255.77: число байт=32 время=44мс TTL=50
Ответ от 5.255.255.77: число байт=32 время=51мс TTL=50

Статистика Ping для 5.255.255.77:
    Пакетов: отправлено = 4, получено = 4, потеряно = 0
    (0% потерь)

Приблизительное время приема-передачи в мс:
    Минимальное = 44мсек, Максимальное = 56 мсек, Среднее = 49 мсек
PS C:\work\zfagaeva> |
```

Рисунок 6: Проверка доступности узла yandex.ru с помощью ping

2.7 Анализирую ICMP-запрос к удалённому узлу

arp or icmp						
No.	Time	Source	Destination	Protocol	Length	Info
9	5.514108	10.90.5.42	5.255.255.77	ICMP	74	Echo (ping) request id=0x0001, seq=5/1280, ttl=128 (reply in 10)
10	5.561182	5.255.255.77	10.90.5.42	ICMP	74	Echo (ping) reply id=0x0001, seq=5/1280, ttl=50 (request in 9)
13	6.523358	10.90.5.42	5.255.255.77	ICMP	74	Echo (ping) request id=0x0001, seq=6/1536, ttl=128 (reply in 14)
14	6.579935	5.255.255.77	10.90.5.42	ICMP	74	Echo (ping) reply id=0x0001, seq=6/1536, ttl=50 (request in 13)
15	7.536714	10.90.5.42	5.255.255.77	ICMP	74	Echo (ping) request id=0x0001, seq=7/1792, ttl=128 (reply in 16)
16	7.680779	5.255.255.77	10.90.5.42	ICMP	74	Echo (ping) reply id=0x0001, seq=7/1792, ttl=50 (request in 15)

> Frame 9: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{3ABB4A4F-DF76-4939-80B7-FC37529...}

Ethernet II, Src: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec), Dst: 72:18:c7:62:fa:43 (72:18:c7:62:fa:43)

- Destination: 72:18:c7:62:fa:43 (72:18:c7:62:fa:43)
 -0 = LG bit: Locally administered address (this is NOT the factory default)
- Source: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)
 -0 = LG bit: Globally unique address (factory default)
 -0 = IG bit: Individual address (unicast)
- Type: IPv4 (0x0800)
- [Stream index: 0]
- Internet Protocol Version 4, Src: 10.90.5.42, Dst: 5.255.255.77
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 68
 - Identification: 0x85bb (34235)
 - > 000. = Flags: 0x0
 - ...0 0000 0000 0000 = Fragment Offset: 0
 - Time to Live: 128
 - Protocol: ICMP (1)
 - Header Checksum: 0x0000 [validation disabled]
 - [Header checksum status: Unverified]
 - Source Address: 10.90.5.42
 - Destination Address: 5.255.255.77
 - [Stream index: 3]
- Internet Control Message Protocol
 - Type: Echo (ping) request (8)
 - Code: 0
 - Checksum: 0x4d56 [correct]
 - [Checksum Status: Good]
 - Identifier (BE): 1 (0x0001)
 - Identifier (LE): 256 (0x0100)
 - Sequence Number (BE): 5 (0x0005)
 - Sequence Number (LE): 1280 (0x0500)
 - [Response frame: 10]
 - > Data (32 bytes)

Рисунок 7: ICMP Echo Request к удалённому узлу yandex.ru

2.8 Анализирую ICMP-ответ удалённого узла

arp or icmp						
No.	Time	Source	Destination	Protocol	Length	Info
9	5.514108	10.90.5.42	5.255.255.77	ICMP	74	Echo (ping) request id=0x0001, seq=5/1280, ttl=128 (reply in 10)
10	5.561182	5.255.255.77	10.90.5.42	ICMP	74	Echo (ping) reply id=0x0001, seq=5/1280, ttl=50 (request in 9)
13	6.523358	10.90.5.42	5.255.255.77	ICMP	74	Echo (ping) request id=0x0001, seq=6/1536, ttl=128 (reply in 14)
14	6.579935	5.255.255.77	10.90.5.42	ICMP	74	Echo (ping) reply id=0x0001, seq=6/1536, ttl=50 (request in 13)
15	7.536714	10.90.5.42	5.255.255.77	ICMP	74	Echo (ping) request id=0x0001, seq=7/1792, ttl=128 (reply in 16)
16	7.600717	5.255.255.77	10.90.5.42	ICMP	74	Echo (ping) reply id=0x0001, seq=7/1792, ttl=50 (request in 15)

> Frame 10: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on Interface \Device\NPF_{3AB84A4F-DF76-4939-8087-FC3752}	0000	fa fe fa	0f 7b ec 72 1
Ethernet II, Src: 72:18:c7:62:fa:43 (72:18:c7:62:fa:43), Dst: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)	0010	00 3c 85 bb	00 00 32 e
Destination: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)	0020	05 2a 00 00	55 56 00 e
.....0..... = LG bit: Globally unique address (factory default)	0030	67 68 69 6a	6b 6c 6d e
.....0..... = IG bit: Individual address (unicast)	0040	77 61 62 63	64 65 66 e
Source: 72:18:c7:62:fa:43 (72:18:c7:62:fa:43)			
.....1..... = LG bit: Locally administered address (this is NOT the factory default)			
.....0..... = IG bit: Individual address (unicast)			
Type: IPv4 (0x0800)			
[Stream index: 0]			
Internet Protocol Version 4, Src: 5.255.255.77, Dst: 10.90.5.42			
0100 = Version: 4			
... 0101 = Header Length: 20 bytes (5)			
> Differentiated Services Field: 0x28 (DSCP: AF11, ECN: Not-ECT)			
Total Length: 60			
Identification: 0x85bb (34235)			
> 0000 = Flags: 0x0			
...0 0000 0000 0000 = Fragment Offset: 0			
Time to Live: 50			
Protocol: ICMP (1)			
Header Checksum: 0xee0d [validation disabled]			
[Header checksum status: Unverified]			
Source Address: 5.255.255.77			
Destination Address: 10.90.5.42			
[Stream index: 3]			
Internet Control Message Protocol			
Type: Echo (ping) reply (0)			
Code: 0			
Checksum: 0x5556 [correct]			
[Checksum Status: Good]			
Identifier (BE): 1 (0x0001)			
Identifier (LE): 256 (0x0100)			
Sequence Number (BE): 5 (0x0005)			
Sequence Number (LE): 1280 (0x0500)			
[Request frame: 9]			
[Response time: 47,074 ms]			
> Data (32 bytes)			

Рисунок 8: ICMP Echo Reply от удалённого узла yandex.ru

3. 3. Анализ протоколов транспортного уровня

3.1 Исследуя HTTP-запрос и TCP

http						
No.	Time	Source	Destination	Protocol	Length	Info
327	10.321988	10.90.5.42	188.184.67.127	HTTP	342	GET /apple-touch-icon.png HTTP/1.1
329	10.564679	188.184.67.127	10.90.5.42	HTTP	414	HTTP/1.1 404 Not Found (text/html)
344	10.745933	10.90.5.42	188.184.67.127	HTTP	497	GET / HTTP/1.1
358	10.922230	188.184.67.127	10.90.5.42	HTTP	932	HTTP/1.1 200 OK (text/html)
366	11.011378	10.90.5.42	188.184.67.127	HTTP	438	GET /favicon.ico HTTP/1.1
383	11.144869	188.184.67.127	10.90.5.42	HTTP	408	HTTP/1.1 200 OK (image/vnd.microsoft.icon)
422	13.871300	10.90.5.42	188.184.67.127	HTTP	557	GET /hypertext/MAN/TheProject.html HTTP/1.1
430	14.062593	188.184.67.127	10.90.5.42	HTTP	1204	HTTP/1.1 200 OK (text/html)
449	17.409536	10.90.5.42	188.184.67.127	HTTP	580	GET /hypertext/MAN/Help.html HTTP/1.1
451	17.958893	188.184.67.127	10.90.5.42	HTTP	1130	HTTP/1.1 200 OK (text/html)

> Frame 449: Packet, 580 bytes on wire (4640 bits), 580 bytes captured (4640 bits) on interface \Device\NPF_{3ABB4A4F-DF76-4939-88B7-F}	0000	72 18 c0 62 fa 43 f8 fe
> Ethernet II, Src: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec), Dst: 72:18:c7:62:fa:43 (72:18:c7:62:fa:43)	0010	82 36 a9 77 40 00 80 06
> Internet Protocol Version 4, Src: 10.90.5.42, Dst: 188.184.67.127	0020	43 7f e6 62 00 50 4a 06
> Transmission Control Protocol, Src Port: 58978, Dst Port: 80, Seq: 1, Ack: 1, Len: 526	0030	02 00 11 e4 00 00 47 45
Source Port: 58978	0040	74 65 78 74 2f 57 57 57
Destination Port: 80	0050	6d 6c 20 48 54 54 50 2f
[Stream Index: 25]	0060	74 3a 20 69 6e 06 6f 2e
[Stream Packet Number: 4]	0070	0a 43 6f 6e 6e 65 63 74
> [Conversation completeness: Complete, WITH_DATA (31)]	0080	70 2d 61 6c 69 76 65 0d
[TCP Segment Len: 526]	0090	2d 49 6e 73 65 63 75 72
Sequence Number: 1 (relative sequence number)	00a0	74 73 3a 20 31 0d 0a 55
Sequence Number (raw): 1241943559	00b0	74 3a 20 4d 6f 7a 69 6c
[Next Sequence Number: 527 (relative sequence number)]	00c0	57 69 6e 64 6f 77 73 20
Acknowledgment Number: 1 (relative ack number)	00d0	20 57 69 6e 36 34 3b 20
Acknowledgment number (raw): 712434136	00e0	6c 65 57 65 62 4b 69 74
0101 = Header Length: 20 bytes (5)	00f0	28 4b 48 54 4d 4c 2c 20
> Flags: 0x018 (PSH, ACK)	0100	6b 6f 29 20 43 68 72 6f
Window: 512	0110	2e 30 2e 30 20 59 61 42
[Calculated window size: 131072]	0120	36 2e 38 2e 30 2e 30 20
[Window size scaling factor: 256]	0130	33 37 2e 33 36 0d 0a 41
Checksum: 8x1e4 [unverified]	0140	65 78 74 2f 68 74 6d 6c
[Checksum Status: Unverified]	0150	74 69 6f 6e 2f 78 68 74
Urgent Pointer: 0	0160	70 70 6c 69 63 61 74 69
> [Timestamps]	0170	3d 30 2e 39 2c 09 6d 61
> [SEQ/ACK analysis]	0180	69 6d 61 67 65 2f 77 65
[Client Contiguous Streams: 1]	0190	2f 61 70 6e 67 2c 2a 2f
[Server Contiguous Streams: 1]	01a0	61 70 70 6c 69 63 61 74
TCP payload (526 bytes)	01b0	65 64 2d 65 78 63 68 61
> Hypertext Transfer Protocol	01c0	3b 71 3d 30 2e 37 0d 0a
	01d0	20 68 74 74 70 3a 2f 2f
	01e0	6e 2e 63 68 2f 68 79 70
	01f0	57 57 2f 54 68 65 50 72
	0200	6d 6c 0d 0a 41 63 63 65
	0210	69 6e 67 3a 20 67 7a 69
	0220	74 65 0d 0a 41 63 63 65
	0230	61 67 65 3a 20 72 75 2c
	0240	0d 0a 0d 0a

Рисунок 9: HTTP GET-запрос и параметры соответствующего TCP-сегмента

3.2 Исследуя HTTP-ответ

http						
No.	Time	Source	Destination	Protocol	Length	Info
327	10.321988	10.90.5.42	188.184.67.127	HTTP	342	GET /apple-touch-icon.png HTTP/1.1
329	10.564679	188.184.67.127	10.90.5.42	HTTP	414	HTTP/1.1 404 Not Found (text/html)
344	10.745933	10.90.5.42	188.184.67.127	HTTP	497	GET / HTTP/1.1
358	10.922230	188.184.67.127	10.90.5.42	HTTP	932	HTTP/1.1 200 OK (text/html)
366	11.011378	10.90.5.42	188.184.67.127	HTTP	438	GET /favicon.ico HTTP/1.1
383	11.144869	188.184.67.127	10.90.5.42	HTTP	408	HTTP/1.1 200 OK (image/vnd.microsoft.icon)
422	13.871300	10.90.5.42	188.184.67.127	HTTP	557	GET /hypertext/WWW/TheProject.html HTTP/1.1
430	14.062593	188.184.67.127	10.90.5.42	HTTP	1204	HTTP/1.1 200 OK (text/html)
449	17.409536	10.90.5.42	188.184.67.127	HTTP	580	GET /hypertext/WWW/Help.html HTTP/1.1
451	17.558893	188.184.67.127	10.90.5.42	HTTP	1130	HTTP/1.1 200 OK (text/html)

> Frame 451: Packet, 1130 bytes on wire (9040 bits), 1130 bytes captured (9040 bits) on interface \Device\NPF_{3A8B4A4F-DF76-4939-8007-000000000000} (0.0.0.0)	0020 05 2a 00 50 e6 62 2a 76
> Ethernet II, Src: 72:18:c7:62:fa:43 (72:18:c7:62:fa:43), Dst: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)	0030 01 f5 8b 0a 00 00 48 54
> Internet Protocol Version 4, Src: 188.184.67.127, Dst: 10.90.5.42	0040 30 30 20 4f 4b 0d 0a 44
> Transmission Control Protocol, Src Port: 80, Dst Port: 58978, Seq: 1, Ack: 527, Len: 1076	0050 2c 20 32 36 20 53 65 70
Source Port: 80	0060 3a 32 30 3a 30 37 20 47
Destination Port: 58978	0070 65 72 3a 20 41 70 61 63
[Stream index: 25]	0080 2d 4d 6f 64 69 66 69 65
[Stream Packet Number: 6]	0090 32 36 20 41 75 67 20 31
> [Conversation completeness: Complete, WITH_DATA (31)]	00a0 33 3a 34 3a 20 47 4d 54
[TCP Segment Len: 1076]	00b0 22 33 34 63 2d 32 38 61
Sequence Number: 1 (relative sequence number)	00c0 30 30 22 0d 0a 41 63 63
Sequence Number (raw): 712434136	00d0 65 73 3a 20 62 79 74 65
[Next Sequence Number: 1077 (relative sequence number)]	00e0 6e 74 2d 4c 65 6e 67 74
Acknowledgment Number: 527 (relative ack number)	00f0 43 6f 0e 6e 65 63 74 69
Acknowledgment number (raw): 1241944085	0100 65 0d 0a 43 6f 6e 74 65
0101 = Header Length: 20 bytes (5)	0110 20 74 65 7b 74 2f 68 74
> Flags: 0x018 (PSH, ACK)	0120 69 74 6c 65 3e 48 65 6c
Window: 501	0130 20 73 6f 66 74 77 61 72
[Calculated window size: 64128]	0140 3e 0a 3c 68 31 3e 48 65
[Window size scaling factor: 128]	0150 57 68 69 63 68 20 70 72
Checksum: 0xb0ba [unverified]	0160 65 20 79 6f 75 20 75 73
[Checksum Status: Unverified]	0170 77 68 69 63 68 20 79 6f
Urgent Pointer: 0	0180 6e 6c 69 6e 65 20 68 65
> [Timestamps]	0190 0a 3c 64 74 3e 3c 61 20
> [SEQ/ACK analysis]	01a0 65 4d 6f 64 65 2f 44 65
[Client Contiguous Streams: 1]	01b0 75 69 63 6b 47 75 69 64
[Server Contiguous Streams: 1]	01c0 69 6e 65 20 4d 6f 64 65
TCP payload (1076 bytes)	01d0 3c 2f 61 3e 0a 3c 64 64
> Hypertext Transfer Protocol	01e0 68 65 20 6c 69 6e 65 20
> Line-based text data: text/html (21 lines)	01f0 77 73 65 72 2c 20 73 65
	0200 6f 70 74 69 6f 6e 20 77
	0210 6d 62 65 72 20 61 6e 64
	0220 45 54 55 52 4e 0a 3c 64
	0230 3c 64 64 3e 46 70 72 20
	0240 56 69 6f 6c 61 2c 20 6a
	0250 73 20 74 68 65 20 71 75
	0260 61 72 6b 20 28 3f 29 20
	0270 72 20 74 68 65 0a 74 6f
	0280 68 61 6e 64 20 63 6f 72
	0290 7e 3c 61 3e 68 73 65 66

Рисунок 10: HTTP-ответ 200 OK и параметры TCP-сегмента сервера

3.3 Анализирую DNS-запрос

dns						
No.	Time	Source	Destination	Protocol	Length	Info
374	11.097691	10.90.5.59	10.90.5.42	DNS	170	Standard query response 0x9974 A sba.yandex.net A 87.250.251.232 A 93.158.134.1
407	13.004119	10.90.5.42	10.90.5.59	DNS	76	Standard query 0xc95a HTTPS home.web.cern.ch
408	13.004342	10.90.5.42	10.90.5.59	DNS	76	Standard query 0xc68d1 A home.web.cern.ch
409	13.257303	10.90.5.59	10.90.5.42	DNS	174	Standard query response 0xc68d1 A home.web.cern.ch CNAME app-catalogue-apps-lb-f
410	13.267124	10.90.5.59	10.90.5.42	DNS	183	Standard query response 0xc95a HTTPS home.web.cern.ch CNAME app-catalogue-apps-
411	13.268073	10.90.5.42	10.90.5.59	DNS	77	Standard query 0x1e56 HTTPS line-mode.cern.ch
412	13.268368	10.90.5.42	10.90.5.59	DNS	77	Standard query 0xc90d A line-mode.cern.ch
413	13.443840	10.90.5.59	10.90.5.42	DNS	174	Standard query response 0x1e56 HTTPS line-mode.cern.ch CNAME lb-paas-line-mode-
414	13.457412	10.90.5.59	10.90.5.42	DNS	194	Standard query response 0xc90d A line-mode.cern.ch CNAME lb-paas-line-mode-4504
409	13.257303	10.90.5.59	10.90.5.42	DNS	174	Standard query response 0xc68d1 A home.web.cern.ch CNAME app-catalogue-apps-lb-f
Frame 408: Packet, 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on interface \Device\NPF_{3ABB4A4F-D776-4939-80B7-FC375} (0000 72 18 c7 62 fa 4 0010 00 3e 8a c0 00 0 0020 05 3b cf 29 00 0 0030 00 00 00 00 00 0 0040 63 65 72 6e 02 0 6)						
> Ethernet II, Src: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec), Dst: 72:18:c7:62:fa:43 (72:18:c7:62:fa:43)						
> Internet Protocol Version 4, Src: 10.90.5.42, Dst: 10.90.5.59						
User Datagram Protocol, Src Port: 53033, Dst Port: 53						
Source Port: 53033						
Destination Port: 53						
Length: 42						
Checksum: 0x1f54 [unverified]						
[Checksum Status: Unverified]						
[Stream index: 27]						
[Stream Packet Number: 1]						
[Timestamps]						
UDP payload (34 bytes)						
Domain Name System (query)						
Transaction ID: 0xc68d1						
Flags: 0x0100 Standard query						
Questions: 1						
Answer RRs: 0						
Authority RRs: 0						
Additional RRs: 0						
Queries						
[Response In: 409]						

Рисунок 11: DNS-запрос имени home.web.cern.ch по протоколу UDP

3.4 Анализирую DNS-ответ

No.	Time	Source	Destination	Protocol	Length	Info
374	11.097691	10.90.5.59	10.90.5.42	DNS	178	Standard query response 0x9974 A sba.yandex.net A 87.250.251.232 A 93.158.134.232 A
407	13.004119	10.90.5.42	10.90.5.59	DNS	76	Standard query 0xc95a HTTPS home.web.cern.ch
408	13.004342	10.90.5.42	10.90.5.59	DNS	76	Standard query 0x68d1 A home.web.cern.ch
409	13.257393	10.90.5.59	10.90.5.42	DNS	174	Standard query response 0x68d1 A home.web.cern.ch CNAME app-catalogue-apps-lb-home.c
410	13.267124	10.90.5.59	10.90.5.42	DNS	183	Standard query response 0xc95a HTTPS home.web.cern.ch CNAME app-catalogue-apps-lb-ho
411	13.268073	10.90.5.42	10.90.5.59	DNS	77	Standard query 0x1e56 HTTPS line-mode.cern.ch
412	13.268368	10.90.5.42	10.90.5.59	DNS	77	Standard query 0xc90d A line-mode.cern.ch
413	13.443840	10.90.5.59	10.90.5.42	DNS	174	Standard query response 0x1e56 HTTPS line-mode.cern.ch CNAME lb-paas-line-mode-45047
414	13.457412	10.90.5.59	10.90.5.42	DNS	194	Standard query response 0xc90d A line-mode.cern.ch CNAME lb-paas-line-mode-45047.cern
427	17.000000	10.90.5.42	10.90.5.59	DNS	00	Standard query response 0xc90d A line-mode.cern.ch CNAME lb-paas-line-mode-45047.cern

> Frame 409: Packet, 174 bytes on wire (1392 bits), 174 bytes captured (1392 bits) on interface \Device\NPF_{3AB8444F-DF76-4939-8087-F	0000	f8 fe 5e 6f 7b ec 72
> Ethernet II, Src: 72:18:c7:62:fa:43 (72:18:c7:62:fa:43), Dst: Intel_E7:7b:ec (f8:fe:5e:6f:7b:ec)	0010	00 a0 e1 f6 40 00 40
> Internet Protocol Version 4, Src: 10.90.5.59, Dst: 10.90.5.42	0020	05 2a 00 35 cf 29 00 00
> User Datagram Protocol, Src Port: 53, Dst Port: 53033	0030	00 02 00 00 00 00 00 04
Source Port: 53	0040	63 65 72 6e 02 63 68 00
Destination Port: 53033	0050	00 01 00 00 00 3c 00 00
Length: 140	0060	61 6c 6f 67 75 65 2d 00
Checksum: 0xcd2f [unverified]	0070	6f 6d 65 04 63 65 72 00
[Checksum Status: Unverified]	0080	2d 43 41 54 41 4c 4f 00
[Stream index: 27]	0090	4c 42 2d 48 4f 4d 45 00
[Stream Packet Number: 2]	00a0	00 01 00 01 00 00 00 00
> [Timestamps]		
UDP payload (132 bytes)		
> Domain Name System (response)		
Transaction ID: 0x68d1		
> Flags: 0x8100 Standard query response, No error		
Questions: 1		
Answer RRs: 2		
Authority RRs: 0		
Additional RRs: 0		
> Queries		
> Answers		
[Request In: 408]		
[Time: 252.961000 milliseconds]		

Рисунок 12: DNS-ответ на запрос имени home.web.cern.ch

3.5 Исследуя протокол QUIC

quic						
No.	Time	Source	Destination	Protocol	Length	Info
506	41.452367	10.90.5.42	173.194.73.95	QUIC	1292	Initial, DCID=3fa715e58f99dfaa, PKN: 1, PADDING, PING, PADDING, PING, PADDING, CRYPT
507	41.452461	10.90.5.42	173.194.73.95	QUIC	1292	Initial, DCID=3fa715e58f99dfaa, PKN: 2, CRYPTO
508	41.452506	10.90.5.42	173.194.73.95	QUIC	1292	Initial, DCID=3fa715e58f99dfaa, PKN: 3, PING, CRYPTO, PADDING, PING, CRYPTO, PADDIN
509	41.452898	10.90.5.42	173.194.73.95	QUIC	124	0-RTT, DCID=3fa715e58f99dfaa
510	41.594901	173.194.73.95	10.90.5.42	QUIC	82	Initial, SCID=ffa715e58f99dfaa, PKN: 1, ACK
Frame 506: Packet, 1292 bytes on wire (10336 bits), 1292 bytes captured (10336 bits) on interface \Device\NPF_{3AB84A4F-DF76-4939-0010-04FE5E774B00} 0000 72 18 c7 62 fa 43 f0						
Ethernet II, Src: Intel_E77Bec (f8fe5e6f7bdec), Dst: 72:18:c7:62:fa:43 (72:18:c7:62:fa:43) 0010 04 fe 5e 77 4b 00 00						
Internet Protocol Version 4, Src: 10.90.5.42, Dst: 173.194.73.95 0020 49 5f 07 f3 01 bb 04						
User Datagram Protocol, Src Port: 61555, Dst Port: 443 0030 3f a7 15 e5 8f 99 df						
Source Port: 61555 0040 0a 30 60 f4 50 0f 3f						
Destination Port: 443 0050 33 03 5f 41 f4 f2 c7						
Length: 1258 0060 94 b7 4f 1f 99 fb 13						
Checksum: 0x0ba1 [unverified] 0070 8e 6d 14 88 19 19 ef						
[Checksum Status: Unverified] 0080 3c 44 89 d6 a4 cf 32						
[Stream index: 37] 0090 4d 4c 58 39 f7 ab cf						
[Stream Packet Number: 1] 00a0 b9 f0 8e 7b 1e a1 1b						
[Timestamps] 00b0 4a 43 bc 29 0e be ff						
UDP payload (1250 bytes) 00c0 4a 2d 1d 0d d2 ae 98						
QUIC IETF 00d0 0d 08 5f fc c5 25 d1						
QUIC Connection information 00e0 46 70 56 87 e9 53 af						
[Packet Length: 1250] 00f0 b0 c2 59 6a 56 4d 0c						
1... = Header Form: Long Header (1) 0100 fb 2f af 18 c2 2a f0						
1.. = Fixed Bit: True 0110 50 03 2b d6 71 da 81						
..00 = Packet Type: Initial (0) 0120 d6 4d 8d c5 71 a9 ec						
[.... 00.. = Reserved: 0] 0130 73 20 0d b2 e0 18 31						
[.... ..00 = Packet Number Length: 1 bytes (0)] 0140 c4 e2 1a 6c bf 66 40						
Version: 1 (0x00000001) 0150 fa 0d ff f8 0e 04 ff						
Destination Connection ID Length: 0 0160 85 53 82 48 11 a3 e2						
Source Connection ID Length: 0 0170 4a c3 f8 62 96 08 70						
Token Length: 70 0180 44 d7 35 00 71 cc c3						
Token: 0010e451690a3060f4500f3f5f3c068c9e98ef54aa33035f41f4f2c7c24217a53bbd81514894874f1f99f81358ca3ece5f9cb2317b8e6d14881919ef 0190 39 9a 61 00 82 2e e6						
Length: 1161 01a0 fd dc 9b 5a 27 5b e9						
[Packet Number: 1] 01b0 c3 fa 17 f3 42 3b f7						
Payload [-]: a4cf320bd1bdf3eb4ea68afa4d4c5839f7abcf47c0f4fc9772bb3c09b9f08e7b1ea11b245402586a178b2e694a43bc290ebe8ff94150ced81c 01c0 fb e0 3e 21 68 0e c1						
PADDING Length: 1 01d0 30 77 d8 aa 2c 7f ff						
PING 01e0 3c fe 61 ed 43 4e a2						
PING 01f0 de b9 0d 1d fa 0e 16						
PING 0200 6c ce 75 6b 0f e6 31						
PING 0210 7c fb 0c 19 8c 85 64						
PING 0220 f6 7f af d2 5f 6b 78						
PING 0230 cc c3 50 ed df 66 ef						
PING 0240 b2 f8 ef b5 5c 73 dc						
PING 0250 c4 ee 0b 3e b8 e0 45						
PING 0260 27 c5 16 26 09 1e a0						
CRYPTO 0270 a5 06 94 88 01 d0 47						
CRYPTO 0280 9a af 7a 4e b5 0c 12						
CRYPTO 0290 9d 37 28 7c a6 f5 5a						
CRYPTO 02a0 4c c6 4d 36 0a d3 1a						

Рисунок 13: Начальный QUIC-пакет поверх UDP к порту 443

3.6 Анализирую QUIC Handshake

No.	Time	Source	Destination	Protocol	Length	Info
518	41.596263	173.194.73.95	10.90.5.42	QUIC	1292	Initial, SCID=ffa715e58f99dfaa, PKN: 3, ACK, PADDING
519	41.598221	173.194.73.95	10.90.5.42	QUIC	1292	Initial, SCID=ffa715e58f99dfaa, PKN: 4, ACK, PADDING
520	41.600240	173.194.73.95	10.90.5.42	QUIC	1292	Initial, SCID=ffa715e58f99dfaa, PKN: 5, CRYPTO, PADDING
521	41.600240	173.194.73.95	10.90.5.42	QUIC	335	Protected Payload (KP0)
522	41.601123	10.90.5.42	173.194.73.95	QUIC	136	Handshake, DCID=ffa715e58f99dfaa
523	41.601682	10.90.5.42	173.194.73.95	QUIC	1288	Protected Payload (KP0), DCID=ffa715e58f99dfaa
524	41.601782	10.90.5.42	173.194.73.95	QUIC	1288	Protected Payload (KP0), DCID=ffa715e58f99dfaa
525	41.601782	10.90.5.42	173.194.73.95	QUIC	1288	Protected Payload (KP0), DCID=ffa715e58f99dfaa

>	Frame 522: Packet, 136 bytes on wire (1088 bits), 136 bytes captured (1088 bits) on interface \Device\NPF_{3AB84A4F-DF76-4939-8087-F}
>	Ethernet II, Src: Intel_6f:7b:ec (8f:fe:5e:6f:7b:ec), Dst: 72:18:c7:62:fa:43 (72:18:c7:62:fa:43)
>	Internet Protocol Version 4, Src: 10.90.5.42, Dst: 173.194.73.95
>	User Datagram Protocol, Src Port: 61555, Dst Port: 443
>	Source Port: 61555
>	Destination Port: 443
>	Length: 102
>	Checksum: 0x071d [unverified]
>	[Checksum Status: Unverified]
>	[Stream index: 37]
>	[Stream Packet Number: 11]
>	[Timestamps]
>	UDP payload (94 bytes)
>	QUIC IETF
>	QUIC Connection Information
>	[Connection Number: 0]
>	[Packet Length: 94]
>	1... = Header Form: Long Header (1)
>	1... = Fixed Bit: True
>	..10 = Packet Type: Handshake (2)
>	Version: 1 (0x00000001)
>	Destination Connection ID Length: 8
>	Destination Connection ID: ffa715e58f99dfaa
>	Source Connection ID Length: 0
>	Length: 77
>	[Expert Info (Warning/Decryption): Failed to create decryption context: Secrets are not available]
>	[Failed to create decryption context: Secrets are not available]
>	[Severity level: Warning]
>	[Group: Decryption]
>	Remaining Payload: 16d7fd91537ccc18c59b1a3336a0687b01a388313793a46c1d667447e5f804a835dc29390371ec7fd9cb9c28f9a2b52b58f7442c80db4

Рисунок 14: QUIC Handshake и переход к защищённому обмену

4. 4. Анализ TCP Handshake

4.1 Исследуя установление TCP-соединения

2	0.06/410	87.250.251.20	10.90.5.42	TCP	66 443 → 58045 [ACK] Seq=1 Ack=2 Win=166 Len=0 SLE=1 SRE=2
7	1.990114	10.90.5.42	188.184.67.127	TCP	66 59343 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
8	1.994338	10.90.5.42	188.184.67.127	TCP	66 54766 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
9	2.083391	188.184.67.127	10.90.5.42	TCP	66 80 → 59343 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1300 SACK_PERM WS=128
10	2.083545	10.90.5.42	188.184.67.127	TCP	54 59343 → 80 [ACK] Seq=1 Ack=1 Win=131072 Len=0
11	2.083933	10.90.5.42	188.184.67.127	HTTP	692 GET /hypertext/WWW/Help.html HTTP/1.1
12	2.089918	188.184.67.127	10.90.5.42	TCP	66 80 → 54766 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1300 SACK_PERM WS=128
13	2.090023	10.90.5.42	188.184.67.127	TCP	54 54766 → 80 [ACK] Seq=1 Ack=1 Win=131072 Len=0
14	2.166351	188.184.67.127	10.90.5.42	TCP	54 80 → 59343 [ACK] Seq=1 Ack=639 Win=63616 Len=0
15	2.167190	188.184.67.127	10.90.5.42	HTTP	250 HTTP/1.1 304 Not Modified
16	2.167190	188.184.67.127	10.90.5.42	TCP	54 80 → 59343 [FIN, ACK] Seq=197 Ack=639 Win=63616 Len=0
17	2.167269	10.90.5.42	188.184.67.127	TCP	54 59343 → 80 [ACK] Seq=639 Ack=198 Win=131072 Len=0
18	2.167805	10.90.5.42	188.184.67.127	TCP	54 59343 → 80 [FIN, ACK] Seq=639 Ack=198 Win=131072 Len=0
19	2.186688	10.90.5.42	87.250.251.20	TLSv1.2	169 Ignored Unknown Record

Рисунок 15: Последовательность TCP SYN, SYN-ACK и ACK при установлении соединения

4.2 Анализирую TCP SYN-ACK

```
Internet Protocol Version 4, Src: 100.104.0.127, Dst: 10.20.3.42
Transmission Control Protocol, Src Port: 80, Dst Port: 59343, Seq: 0, Ack: 1, Len: 0
  Source Port: 80
  Destination Port: 59343
  [Stream index: 1]
  [Stream Packet Number: 2]
  > [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 0]
  Sequence Number: 0 (relative sequence number)
  Sequence Number (raw): 2615809941
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 2854257989
  1000 .... = Header Length: 32 bytes (8)
  ▾ Flags: 0x012 (SYN, ACK)
    000. .... = Reserved: Not set
    ...0 .... = Accurate ECN: Not set
    ....0... = Congestion Window Reduced: Not set
    ....0.. = ECN-Echo: Not set
    ....0. = Urgent: Not set
    ....1... = Acknowledgment: Set
    ....0... = Push: Not set
    ....0.. = Reset: Not set
  ▾ ....1... = Syn: Set
    ▾ [Expert Info (Chat/Sequence): Connection establish acknowledge (SYN+ACK): server port 80]
      [Connection establish acknowledge (SYN+ACK): server port 80]
      [Severity level: Chat]
      [Group: Sequence]
      ....0... = Fin: Not set
      [TCP Flags: .....A..S.]
  Window: 64240
  [Calculated window size: 64240]
  Checksum: 0x99ef [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  > Options: (12 bytes), Maximum segment size, No-Operation (NOP), No-Operation (NOP), SACK permitted, No-Operation (NOP), Window sca
  > [Timestamps]
  > [SEQ/ACK analysis]
  [Client Contiguous Streams: 1]
  [Server Contiguous Streams: 1]
```

Рисунок 16: Подробный анализ флагов и параметров TCP SYN-ACK

4.3 Строю график потока TCP



Рисунок 17: График потока TCP и HTTP в Wireshark

4.4 Анализирую изменение Seq и Ack

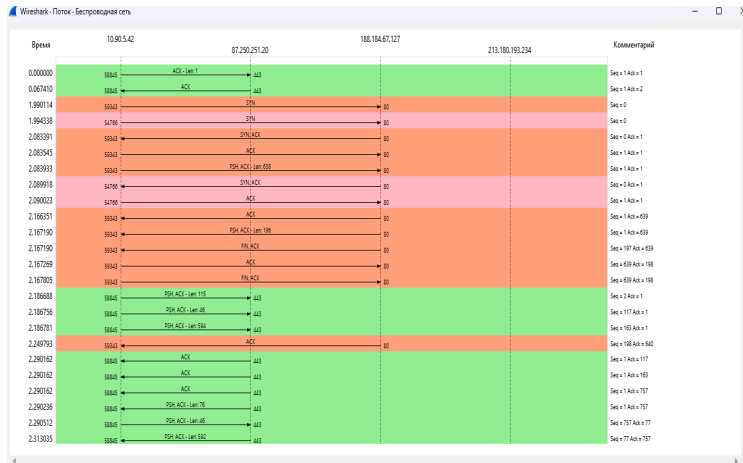


Рисунок 18: Изменение Seq, Ack и управляющих флагов на графике TCP-потока

5. 5. Выводы по проделанной работе

5.1 Вывод

В ходе лабораторной работы были изучены параметры сетевых интерфейсов и структура MAC-адресов.

С помощью Wireshark был выполнен захват и анализ кадров протоколов **ARP** и **ICMP**.

Были определены MAC- и IP-адреса источников и получателей сетевых пакетов.

Проанализирована работа протоколов **HTTP** и **TCP**, а также передача DNS-запросов и ответов по протоколу **UDP**.

Был исследован протокол **QUIC**, работающий поверх UDP и использующий защищённое установление соединения.

Также был подробно рассмотрен процесс TCP handshake, состоящий из последовательности сообщений **SYN**, **SYN-ACK** и **ACK**.

В результате были получены практические навыки анализа сетевого трафика и структуры протоколов различных уровней модели сети.